

Neea Rusch

Curriculum vitae

✉ neea.rusch@it.uu.se
🔗 neea.pl
🆔 0000-0002-7354-5330
📄 dblp.org/pid/296/3722
🐙 github.com/nkrusch

Research Interests

Programming languages, formal methods, implicit computational complexity, static program analysis.

Education

- 8/2021–12/2025 Doctor of Philosophy, Computer & Cyber Sciences, *Augusta University*, Augusta, GA USA.
- 8/2014–5/2016 Master of Science, Computer Science, *Georgia Institute of Technology*, Atlanta, GA USA.
- 8/2011–5/2014 Bachelor of Science, Computer Science, *Augusta University*, Augusta, GA USA.

Professional Appointments & Employment

- 10/2025–present 🇸🇪 Postdoctoral Researcher, *Department Information Technology, Uppsala University*, Uppsala, Sweden.
- 8/2025–10/2025 🇺🇸 Graduate Research Assistant, *The Graduate School, Augusta University*, Augusta GA, USA.
- 8/2021–5/2024 🇺🇸 Graduate Research Assistant, *The Graduate School, Augusta University*, Augusta GA, USA.
- 7/2020–1/2021 🇺🇸 Senior Consultant, *Sparq (formerly Rural Sourcing) Inc.*, Augusta GA, USA.
- 1/2019–12/2021 🇺🇸 Instructor, *School of Computer & Cyber Sciences, Augusta University*, Augusta GA, USA.
- 8/2018–5/2019 🇺🇸 Android Developer, *TaxSlayer*, Augusta GA, USA.
- 5/2014–8/2018 🇺🇸 Software Developer, *TaxSlayer*, Augusta GA, USA.

Publications

- 2025 Neea Rusch. “Applied Implicit Computational Complexity”. PhD thesis. Augusta, Georgia, United States: Augusta University, Sept. 2025. DOI: 10.5281/zenodo.17148449.
- 2023 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “Distributing and Parallelizing Non-canonical Loops”. In: *Verification, Model Checking, and Abstract Interpretation*. Springer Nature Switzerland, 2023, pp. 1–24. DOI: 10.1007/978-3-031-24950-1_1.
- 2023 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “pymwp: A Static Analyzer Determining Polynomial Growth Bounds”. In: *Automated Technology for Verification and Analysis*. Springer International Publishing, 2023, pp. 263–275. DOI: 10.1007/978-3-031-45332-8_14.
- 2022 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “mwp-Analysis Improvement and Implementation: Realizing Implicit Computational Complexity”. In: *7th International Conference on Formal Structures for Computation and Deduction (FSCD 2022)*. Vol. 228. LIPIcs. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2022, 26:1–26:23. DOI: 10.4230/LIPIcs.FSCD.2022.26.
- 2022 Neea Rusch. “Formally Verified Resource Bounds Through Implicit Computational Complexity”. In: *Companion Proceedings of the 2022 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity*. SPLASH Companion 2022. ACM, 2022, pp. 17–20. DOI: 10.1145/3563768.3565545.

Software

- 2024 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. *pymwp: MWP analysis in Python*. Version 0.5.2. Nov. 2024. DOI: 10.5281/zenodo.7879822.
- 2022 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. *Distributing and Parallelizing Non-canonical Loops – Artifact*. Version 1.0. Sept. 2022. DOI: 10.5281/zenodo.7080145.

Research Presentations

Presentation titles and slides are available at neea.pl/posts/talks.

- 2025-07-03 🇳🇴 Doctoral Symposium at ECOOP 2025, *Høgskulen på Vestlandet, Bergen, Norway*.
- 2025-06-03 🇸🇪 Programming Languages Seminar, *Uppsala University, Uppsala, Sweden*.
- 2024-10-14 🇺🇸 The 19th Workshop on Programming Languages and Analysis for Security (PLAS'24), *Salt Lake City UT, USA*.
- 2024-08-20 🇫🇮 Theory Seminar, *Aalto University, Espoo, Finland*.
- 2024-08-07 🇩🇪 Lightning talk at the Marktoberdorf Summer School, *Herrsching, Germany*.
- 2024-07-10 🇳🇱 5-min. talks at the Computer Security Foundations Symposium, *University of Twente, Enschede, the Netherlands*.
- 2024-04-23 🇺🇸 The 3-Minute Thesis (3MT), research communication competition, *Augusta University, Augusta GA, USA*.
- 2023-10-25 🇸🇬 The 21st International Symposium on Automated Technology for Verification and Analysis (ATVA'23), *Singapore Management University, Singapore*.
- 2023-10-20 🇭🇰 The International Conference on Security and Privacy in Communication Networks (SecureComm'23), *Hong Kong Polytechnic University, Hong Kong SAR*.
- 2023-09-15 🇩🇪 Lightning talk at the International Autumn School on Proof and Computation, *Herrsching, Germany*.
- 2023-03-10 🌐 SCOT Webinar: A Seminar on Semantic and Formal Approaches to Complexity, *online*.
- 2023-01-21 🇺🇸 The Ninth International Workshop on Coq for Programming Languages (CoqPL'23), *Boston MA, USA*.
- 2023-01-16 🇺🇸 The 24th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI'23), *Boston MA, USA*.
- 2022-12-06 🇳🇿 The 2022 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (SPLASH'22) Doctoral Symposium, *University of Auckland, New Zealand*.
- 2022-06-20 🇫🇷 The 28th International Conference on Types for Proofs and Programs (TYPES'22), *University of Nantes, France*.
- 2022-04-01 🇺🇸 The 37th Annual Graduate Research Day – poster, *Augusta University, USA*.
- 2021-11-18 🇫🇷 Laboratoire d'Informatique de Paris Nord (LIPN) seminar, *Université Sorbonne Paris Nord, France*.
- 2021-11-15 🇫🇷 Inria Celtique research team seminar, *IRISA Rennes, France*.
- 2021-03-26 🌐 The 36th Annual Graduate Research Day, *online, Augusta University, USA*.

Professional Services & Reviews

PC Chair **Artifact Evaluation.**

FormaliSE – 2027.

PC Member **Artifact Evaluation Committee.**

APLAS – Asian Symposium on Programming Languages and Systems – 2022.

CAV – Computer Aided Verification – 2024, 2025.

CCS – Computer and Communications Security – 2026.

COORDINATION – International Conference on Coordination Models and Languages – 2023.

ECOOP – European Conference on Object-Oriented Programming – 2023, 2024, 2025.

ICFP – International Conference on Functional Programming – 2022, 2023, 2024, 2025, 2026.

ICSE – International Conference on Software Engineering – 2024, 2027.

iFM – Integrated Formal Methods – 2025.

OOPSLA – Object-oriented Programming, Systems, Languages, and Applications – 2024, 2025, 2026.

PLDI – Programming Language Design and Implementation – 2022, 2023.

POPL – Principles of Programming Languages – 2024, 2025, 2026, 2027.

PPoPP – Principles and Practice of Parallel Programming – 2023.

SOSP – Symposium on Operating Systems Principles – 2023.

TACAS – Tools and Algorithms for the Construction and Analysis of Systems – 2025, 2026.

⟨Programming⟩ – Art, Science, and Engineering of Programming – 2025.

- PC Member **Program Committee.**
 ICE – Interaction and Concurrency Experiences workshop – 2026.
 ECOOP Research Papers, Extended Review Committee – 2023, 2024.
- Editor **Journal Editor.**
 JOSS – The Journal of Open Source Software – 2025–present.
- Reviewer **Journal Reviews.**
 JOSS – The Journal of Open Source Software – 2023–2026.
 Sci. Comput. Program. – Science of Computer Programming – 2024.
- Organization Committee.**
- Organizer SERPL – The Southeast Regional Programming Languages Seminar – 2023.
 VerifyThis – competition co-organizer – 2027.
- Volunteer Student volunteer at POPL’25, ETAPS’24, CAV’23, POPL’23, and SPLASH’22.
- Website chair CAV – Computer Aided Verification – 2026.

Teaching Experiences

Details at neea.pl/posts/teaching.

Course Instruction (9x)

Graduate/Master level.

- 2026/27 🇸🇪 (scheduled) Software Testing, *instructor*, Uppsala University.

Undergraduate/Bachelor level.

- 2019–2021 🇺🇸 Principles of Computer Programming I (CSCI 1301), *instructor*, Augusta University.
 Total enrollment 228, taught 6 times.
- 2019 🇺🇸 Introduction to Computers and Programming (CSCI 1200), *instructor*, Augusta University.
 Total enrollment 24, taught 2 times.

Student Project Supervision

- 2026 🇸🇪 Bachelor’s Project, Computer Science, *supervisor*, Uppsala University.
- 2026 🇸🇪 Master’s Thesis (3x), Computer Science and Data Science, *subject reviewer*, Uppsala University.
- 2023 🇺🇸 Senior Capstone Project, Computer Science, *co-mentor*, Augusta University.

Academic Awards, Fellowships & Scholarships

Details at neea.pl/awards ◦ total monetary awards over US\$ 30,000.

- 10/2026 Distinguished Artifact Reviewer of OOPSLA 2026, *by the AEC chairs*.
- 4/2026 Distinguished Artifact Reviewer of TACAS 2026, *by the AEC chairs and the ETAPS Steering Committee*.
- 10/2025 Distinguished Artifact Reviewer of OOPSLA 2025, *by the AEC chairs*.
- 7/2025 Distinguished Artifact Reviewer of ECOOP 2025, *by the AEC chairs*.
- 5/2025 Distinguished Artifact Reviewer of TACAS 2025, *awarded by the AEC chairs*.
- 10/2024 Distinguished Artifact Reviewer of OOPSLA 2024, *awarded by the AEC chairs*.
- 4/2024 3 Minute Thesis, Contest Finalist, *organized by The Graduate School at Augusta University*.
- 1/2024 Ralph W. Shrader Graduate Diversity Scholarship, *by AFCEA Educational Foundation*.
- 1/2023 Student Research Program Award, *by Augusta University Provost’s Office and Medical College of Georgia*.
- 4/2022 Excellence in Research in Computer & Cyber Sciences, *by The Graduate School at Augusta University*.
- 8/2015 Grace Hopper Scholarship, *by College of Computing, Georgia Institute of Technology*.
- 4/2014 The Outstanding Computer Science Student Award, Class of 2014, *by Augusta University*.
- 5/2013 Savannah River Scholars Program Scholarship, *by Augusta University and funded by NSF*.

Schools, Seminars & Workshops

- 9/2026  Challenges in Future Software Stack Development for AI/HPC, Saint-Hippolyte, France, Sep 28–Oct 2, 2026.
- 5/2026  Making Replications Count Hackathon, Universität Münster, Germany, May 4–6, 2026.
- 4/2026  VerifyThis 2026, verification competition, Turin, Italy, Apr 11–12, 2026.
- 5/2025  Programming Language Implementation Summer School, Bertinoro, Italy, May 26–31, 2025.
- 8/2024  Marktoberdorf Summer School, Herrsching, Germany, Aug 6–17, 2024.
- 4/2024  VerifyThis 2024, verification competition, Luxembourg City, Luxembourg, Apr 6–7, 2024.
- 9/2023  International Autumn School on Proof and Computation, Herrsching, Germany, Sep 10–16, 2023.
- 7/2023  Verification Mentoring Workshop (VMW) at CAV 2023, Paris, France, Jul 18, 2023.
- 5/2023  Twelfth Summer School on Formal Techniques (SSFT 2023) & the First Formal Methods in the Field Bootcamp, on principles and practice of formal techniques, Atherton CA, USA, May 24–Jun 2, 2023.
- 6/2022  Programming Languages Mentoring Workshop, PLMW @ PLDI 2022, San Diego CA, USA, Jun 13–14, 2022.
- 11/2021  Dagstuhl Seminar 21453, on Static Analyses of Program Flows: Types and Certificate for Complexity, Schloss Dagstuhl, Germany, Nov 7–12, 2021.
- 6/2021  Oregon Programming Languages Summer School (OPLSS 2021), on foundations of programming and security, online, Jun 14–26, 2021.
- 4/2021  Midlands Graduate School, in the foundations of computing science, online, Apr 12–16, 2021.

University Service

- 1/2023–5/2025 Organizer of $\Delta\Delta\Delta$'s weekly Programming Languages Reading Group, *Augusta University*.
- 8/2021–5/2025 Member of The Graduate School's Graduate Student Council (TGS-GSC), *Augusta University*.
- 9/2023–5/2025 Representative in the Graduate Student Government Association (GSGA), *Augusta University*.
- 5/2023–8/2024 Vice President and co-founder of $\Delta\Delta\Delta$ student organization, *Augusta University*.
- 9/2023–5/2024 GSGA Funding & Conferences Database Ad Hoc Committee, *Augusta University*.
- 8/2024–5/2025 GSGA Event Planning Committee, *Augusta University*.

Languages

English, Finnish